

SKILLPULSE: SERVERLESS DEPLOYMENT GUIDE

1. Introduction & Serverless Architecture

SkillPulse is a modern skill-tracking application built to log study sessions and visualize learning progress. While the standard course deploy pipeline targets an Azure VM using a self-hosted agent, this guide explains how to run the entire stack for free over the internet using a fully serverless approach.

- * **Frontend:** Static HTML, CSS, and JS files hosted on Vercel's Edge network for fast global performance.
- * **Backend:** Go (REST API using Gin framework) running inside Vercel Serverless Functions.
- * **Database:** MySQL-compatible database hosted on TiDB Cloud Serverless (Starter plan) with TLS/SSL encryption.

2. Database Setup on TiDB Cloud

First, we set up a cloud-hosted MySQL database. TiDB Cloud offers a Starter plan that is permanently free and comes pre-configured for public TLS connections.

- * **Step A:** Log in to the TiDB Cloud Console (tidbcloud.com) and click 'Create Resource'.
- * **Step B:** Select the 'Starter' plan (\$0/month) and set the Region closest to you (e.g., ap-southeast-1 AWS).
- * **Step C:** In the 'Connect' modal, copy the Host, Username (prefix.root), and generated Password. Keep the password safe.
- * **Step D:** Ensure the IP Access List is set to allow connections from anywhere by setting the firewall rule to allow '0.0.0.0' to '255.255.255.255'. This is required because Vercel uses dynamic IP addresses.
- * **Step E:** Go to the 'SQL Editor' tab, select the default pre-provisioned 'test' database, paste the schema initialization script, and click 'Run' to create the 'skills' and 'learning_logs' tables.

3. Configuring Vercel Environment Variables

For security reasons, database secrets are not stored in the repository. Instead, they are injected directly into Vercel's runtime environment.

Configure these keys under Vercel Settings > Environment Variables:

```
--- Vercel Environment Variables ---
DB_HOST      = gateway01.ap-southeast-1.prod.aws.tidbcloud.com
DB_PORT      = 4000
DB_USER      = AoZTmPfpCkoRyoV.root (Your generated username)
DB_PASSWORD  = [Your generated database password]
DB_NAME      = test
DB_SSL       = skip-verify
```

4. Project Configuration for Vercel Go Runtime

Vercel requires the Go API to be structured as a Serverless function. We define this in the `vercel.json` configuration file at the root of the repository, mapping requests under `/api` to the API function.

```
--- vercel.json Configuration ---
{
  "version": 2,
  "builds": [
    { "src": "api/index.go", "use": "@vercel/go" },
    { "src": "frontend/**", "use": "@vercel/static" }
  ],
  "rewrites": [
    { "source": "/api/(.*)", "destination": "api/index.go" },
    { "source": "/health", "destination": "api/index.go" },
    { "source": "/css/(.*)", "destination": "frontend/css/$1" },
    { "source": "/js/(.*)", "destination": "frontend/js/$1" },
    { "source": "/", "destination": "frontend/index.html" },
    { "source": "/index.html", "destination": "frontend/index.html" }
  ]
}
```

5. How to Deploy (Step by Step)

- * **Step 1:** Clone the repository to your local system and check out the main branch.
- * **Step 2:** Log in to your Vercel Account and click 'Add New' > 'Project'.
- * **Step 3:** Import your GitHub repository containing the SkillPulse codebase.
- * **Step 4:** Open the 'Environment Variables' accordion menu and input the `DB_*` environment keys shown in Section 3.
- * **Step 5:** Click 'Deploy'. Vercel will compile the Go handler binary and deploy the static assets.
- * **Step 6:** Verify connection: Open the deployment URL and refresh the page. The app will fetch skills successfully!

6. Key Architecture Benefits

- * **Serverless Go:** Zero maintenance server running Go API on-demand, scaling down to zero when inactive.
- * **TiDB Serverless:** An autoscaling database that maintains full MySQL compatibility and ACID safety.
- * **No Running Cost:** Both Vercel Hobby tier and TiDB Cloud Starter plan are 100% free for developers.